Lecture 5: Practice Problems — Answers and Hints

Math for Machine Learning

October 24, 2026

Ordinary Differential Equations

1.1. (a) $x(t) = 3e^{-2t}$, $x(1) = 3e^{-2} \approx 0.406$; it halves after $\frac{\log 2}{2} \approx 0.347$. (b) With $u = 4e^{-t}$: $x' = \frac{u}{(1+u)^2}$ and $x(1-x) = \frac{1}{1+u} \cdot \frac{u}{1+u}$, which agree, and $x(0) = \frac{1}{5}$. So $x(t) = \sigma(t - \log 4)$, $x(2) \approx 0.649$, and $x(t) \to 1$.

1.2. (a) $x' = \frac{1}{(1-t)^2} = x^2$ and $x(0) = 1$. The solution tends to ∞ as $t \to 1$: it blows up in finite time. The theorem does not apply, since $F(x) = x^2$ is not Lipschitz continuous on $\mathbb{R}$: $|x^2 - y^2| = |x+y||x-y|$, and $|x+y|$ is unbounded. (b) For $x = t^2$: $x' = 2t = 2\sqrt{t^2}$. The function $2\sqrt{|x|}$ is not Lipschitz continuous near 0, where its slope is unbounded, so uniqueness can fail.

Linear Systems

2.1. (a) Eigenvalue 1 with eigenvector $(1,1)$, and -3 with $(1,-1)$; $(1,0) = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$, so $\mathbf{x}(t) = \frac{1}{2}e^{t}(1,1)^\top + \frac{1}{2}e^{-3t}(1,-1)^\top$ and $\mathbf{x}(1) \approx (1.384, 1.334)^\top$. (b) Exactly the multiples of $(1,-1)$, i.e., the line $x_2 = -x_1$; all other solutions grow along $(1,1)$. (c) $e^{At} = \frac{1}{2}\begin{pmatrix} e^t + e^{-3t} & e^t - e^{-3t} \\ e^t - e^{-3t} & e^t + e^{-3t} \end{pmatrix}$.

2.2. (a) $\frac{d}{dt}\|\mathbf{x}\|^2 = 2\mathbf{x}^\top A\mathbf{x} = 0$, and e^{At} is the rotation by the angle ωt, so the period is $\frac{2\pi}{\omega}$. (b) $\|\mathbf{x} + hA\mathbf{x}\|^2 = \|\mathbf{x}\|^2 + 2h\mathbf{x}^\top A\mathbf{x} + h^2\|A\mathbf{x}\|^2 = (1 + h^2\omega^2)\|\mathbf{x}\|^2$. One period takes $\frac{2\pi}{0.1} \approx 63$ steps, so the norm grows by $(1.01)^{31.4} \approx 1.37$.

Gradient Flow and Euler's Method

3.1. (a) $x_1' = -2x_1$ and $x_2' = -8x_2$, so $\mathbf{x}(t) = (2e^{-2t}, e^{-8t})$ and $f(\mathbf{x}(t)) = 4e^{-4t} + 4e^{-16t}$, which decreases. (b) $\mathbf{x}_{k+1} = \mathbf{x}_k - h\nabla f(\mathbf{x}_k) = \big((1-2h)x_{1,k}, (1-8h)x_{2,k}\big)$, which is gradient descent with $\eta = h$. It converges if $|1-2h| < 1$ and $|1-8h| < 1$, i.e., $0 < h < 0.25$. For $h = 0.2$, the factors are 0.6 and -0.6 (convergence with oscillation in x_2); for $h = 0.3$, x_2 is multiplied by -1.4 and diverges, although the exact flow converges.

3.2. $0.75^4 \approx 0.316$ and $0.875^8 \approx 0.344$, compared with $e^{-1} \approx 0.368$; the errors 0.051 and 0.024 differ by a factor of about 2, as expected for a method with error proportional to h.

Flows, Volumes and Densities

4.1. (a) $\operatorname{div} F = \operatorname{tr} A = -2$, so areas are multiplied by e^{-2t}. The eigenvalues are 1 and -3 (triangular matrix), and $\det e^{At} = e^t e^{-3t} = e^{-2t}$. (b) $DF = \operatorname{diag}(1 - 3x^2, -1)$, so $\operatorname{div} F = -3x^2$.

$F = \mathbf{0}$ at $(0,0)$ (divergence 0) and $(\pm 1, 0)$ (divergence -3). Areas are preserved to first order on the line $x = 0$ and compressed everywhere else.

4.2. (a) $P(x(t) \le a) = P(x(0) \le e^t a)$; differentiating in a gives $e^t \varphi(e^t a)$. (b) With $\varphi'(u) = -u\varphi(u)$: $\frac{\partial p_t}{\partial t} = e^t \varphi(e^t x) + e^{2t} x\, \varphi'(e^t x)$, and $-\frac{\partial}{\partial x}(-x p_t) = p_t + x \frac{\partial p_t}{\partial x} = e^t \varphi(e^t x) + x e^{2t} \varphi'(e^t x)$. (c) $\log p_t(x(t)) = t + \log \varphi(x(0))$, which increases at the rate 1. The flow contracts lengths by e^{-t} (here $\operatorname{div} F = -1$), so the same probability is squeezed into a smaller interval, and the density increases.

Stochastic Differential Equations

5.1. (a) $\mathcal{N}(0, 3)$; $B(4) \sim \mathcal{N}(0, 4)$, so $P(B(4) > 2) = P(Z > 1) \approx 0.159$. (b) Cov $= \min(1, 2) = 1$, so Corr $= \frac{1}{\sqrt{2}} \approx 0.707$. $B(1) + B(2) = 2B(1) + (B(2) - B(1))$ is a sum of independent Gaussians, $\mathcal{N}(0, 4{+}1) = \mathcal{N}(0, 5)$. (c) $B(2) = 1 + (B(2) - B(1)) \sim \mathcal{N}(1, 1)$, since the increment is independent of $B(1)$. The vector $(B(1), B(2))$ has the covariance matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, so $B(1) \mid B(2) = 2$ is $\mathcal{N}(\frac{1}{2} \cdot 2,\ 1 - \frac{1}{2}) = \mathcal{N}(1, \frac{1}{2})$: the mean lies on the straight line from $(0, 0)$ to $(2, 2)$ (a "Brownian bridge").

5.2. (a) $X_1 = 1 - 0.2 + 0.5\sqrt{0.1} \cdot 0.5 \approx 0.879$ and $X_2 = 0.8 \cdot 0.879 - 0.5\sqrt{0.1} \approx 0.545$. (b) $\mathbb{E}[X_{k+1}] = 0.8\, \mathbb{E}[X_k]$, so $\mathbb{E}[X_k] = 0.8^k$, and $\mathbb{E}[X_{10}] \approx 0.107$, compared with $e^{-2} \approx 0.135$: the error of Euler's method for $x' = -2x$. (c) The SDE: $\frac{\sigma^2}{2\theta} = \frac{0.25}{4} = 0.0625$. The chain: $\frac{\sigma^2}{\theta(2 - \theta h)} = \frac{0.25}{3.6} \approx 0.0694$. A stationary distribution exists if $|1 - 2h| < 1$, i.e., $0 < h < 1$.

5.3. The variance is $\sigma^2 h^{2\gamma} \cdot \frac{t}{h} = \sigma^2 t\, h^{2\gamma - 1}$. For $\gamma > \frac{1}{2}$, the noise disappears in the limit (we get an ODE); for $\gamma < \frac{1}{2}$, it explodes. Only $\gamma = \frac{1}{2}$ gives a limit with variance $\sigma^2 t$, independently of h, which is why Brownian motion has $\operatorname{Var}(B(t)) = t$ and the Euler–Maruyama noise is scaled by $\sqrt{h}$.

5.4. (a) The increments are $\sqrt{h}\, Z_k$ with i.i.d. $Z_k \sim \mathcal{N}(0, 1)$, and $h Z_k^2$ has the mean h and the variance $h^2(\mathbb{E}[Z^4] - 1) = 2h^2$. Summing n independent terms gives $\mathbb{E}[Q_n] = nh = T$ and $\operatorname{Var}(Q_n) = 2nh^2 = 2Th \to 0$, so $Q_n \to T$ by Chebyshev's inequality. (b) $|f((k+1)h) - f(kh)| = |f'(\xi_k)| h \le Ch$ with $C = \max |f'|$, so the sum is at most $nC^2h^2 = C^2 Th \to 0$. Since $Q_n \to T > 0$, Brownian paths cannot be continuously differentiable.

Langevin Dynamics and Diffusion Models

6.1. (a) $x_{k+1} = x_k - \frac{h}{s^2}(x_k - \mu) + \sqrt{2h}\, z_k$, so $a = 1 - \frac{h}{s^2}$. (b) If $x_k - \mu \sim \mathcal{N}(0, v)$, then $x_{k+1} - \mu \sim \mathcal{N}(0, a^2 v + 2h)$, and $v = a^2 v + 2h$ gives $v = \frac{2h}{1 - a^2} = \frac{2h}{2h/s^2 - h^2/s^4} = \frac{2s^2}{2 - h/s^2}$; the chain converges to it if $|a| < 1$, i.e., $0 < h < 2s^2$. The values give $\frac{0.5}{1.96} \approx 0.2551$ instead of 0.25; the bias vanishes as $h \to 0$. (c) Gradient descent on U, which converges to the minimizer μ. For a posterior, the minimizer of $U = -\log p(\theta \mid \mathcal{D})$ is the MAP estimate; the noise turns optimization into sampling.

6.2. (a) x_t is Gaussian as a sum of independent Gaussians. The mean is multiplied by $\sqrt{1 - \beta}$ in each step, and the variance satisfies $v_t = (1 - \beta) v_{t-1} + \beta$; if $v_{t-1} = 1 - (1 - \beta)^{t-1}$, then $v_t = (1 - \beta) - (1 - \beta)^t + \beta = 1 - (1 - \beta)^t$. So $x_t = (1 - \beta)^{t/2} x_0 + \sqrt{1 - (1 - \beta)^t}\, z$ can be sampled in one step, without simulating the chain. (b) $t = 10$: mean 2.71, variance 0.183; $t = 100$: 1.09 and 0.867; $t = 500$: 0.019 and 1.000. For large t, x_t is approximately $\mathcal{N}(0, 1)$ for every data point; generation starts from this distribution and runs the process backwards. (c) Euler–Maruyama gives $x_{k+1} = x_k - \frac{h}{2} x_k + \sqrt{h}\, z_k = (1 - \frac{\beta}{2}) x_k + \sqrt{\beta}\, z_k$ for $h = \beta$. Its stationary variance $\frac{\sigma^2}{2\theta} = 1$ agrees with (b).